SOLUTION
SECTION-I

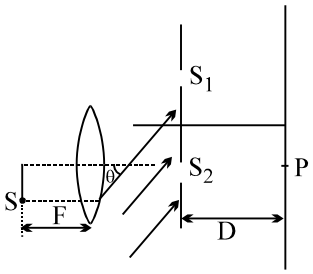
Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (A)

Sol. $\tan\theta = \theta = \sin\theta = \dfrac{d/2}{F}$

suppose P is max.

then $d\sin\theta = S_1P - S_2P = \dfrac{dy}{D}$



$\therefore \dfrac{dD}{2F} = y$ **Ans.**]

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2. Ans. (A)

Sol. Phase difference between waves through S_1 and S_2 is $2 \times (\lambda/6) \times (2\pi/\lambda) = 2\pi/3$ and that through S_1 and S_3 is 2. $2\lambda/3.2\pi/\lambda = 8\pi/3$

$y = A\sin\theta + A\sin(\theta - 2\pi/3) + A\sin(\theta - 8\pi/3)$

$\therefore$ Resultant amplitude $= \sqrt{3}A$

As, Intensity $\propto$ (Amplitude)^2, new intensity $= 3I$

3. Ans. (D)

Sol. Radius of curvature f

4. Ans. (A)

Sol. On comparing optical path we get

$$t\mu_0\left(1-\frac{Z^2}{Z_0}\right) + \sqrt{Z^2+F^2} = t\mu_0 + F$$

$$\Rightarrow \sqrt{Z^2+F^2} = \frac{\mu_0 t Z^2}{Z_0^2} + F$$

$$\Rightarrow Z^2 + F^2 = \mu_0^2 t^2\left(\frac{Z^2}{Z_0^2}\right)^2 + F^2 + 2\mu_0 t\frac{Z^2}{Z_0^2}F$$

since t is very small

$$\Rightarrow Z^2 = 2\mu_0 t\frac{Z^2}{Z_0^2}F \qquad \Rightarrow F = Z_0^2/(2\mu_0 t)$$

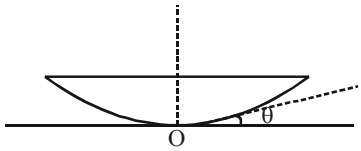
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5. Ans. (B)

Sol. No. of fringes shift (N) $= \frac{t}{\lambda}(\mu-1) = \frac{(1.6-1)\times 1.8\times 10^{-5}}{900\times 10^{-9}} = 12$

6. Ans. (A)

Sol. $\beta = \frac{\lambda}{2\mu\theta}$



$\beta = \frac{\lambda}{2\theta}$

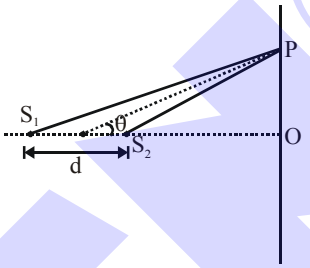
θ increases as we move from point O. Hence β decreases.

7. Ans. (C)

Sol. $\Delta x = S_1P - S_2P = d\cos\theta$

$d\cos\theta = n\lambda$ (maxima)

$d\cos\theta = \left(n+\frac{1}{2}\right)\lambda$ (minima)



$\cos\theta = \frac{\left(n+\frac{1}{2}\right)\lambda}{d}$

For $d = 4.3\ \lambda$

$\cos\theta = \frac{\left(n+\frac{1}{2}\right)\lambda}{4.3\lambda} = \frac{\left(n+\frac{1}{2}\right)}{4.3}$

$n = 0, 1, 2, 3$

4 minimas above 0 & 4 minimas below 0

$\therefore$ Total no. of minimas is 8

At 0, $\theta = 0°$, $d = n\lambda$ (maxima)

At 0, $\theta = 0°$, $d = \left(n+\frac{1}{2}\right)\lambda$ (minima)

8. Ans. (A)

Multiple Correct Answer Type — 8 Q. [4 M (–1)]

9. Ans. (A,B,D)

Sol. From 2nd graph, when $\theta = \frac{\pi}{2}$, $\phi = 4\pi = 2\lambda$ = distance between S_1 & S_2

At $\theta = \frac{\pi}{6}$, $\phi = d\sin\theta = 2\lambda\frac{\ell}{2} = \lambda$.

$\therefore$ Intensity is maximum.

10. Ans. (B,D)

Sol. $\left(\sqrt{I_1}+\sqrt{I_2}\right) = I_{max}$

$I_{min} = \left(\sqrt{I_1}-\sqrt{I_2}\right)^2$

$\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$

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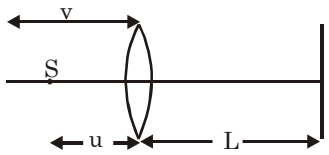
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11. Ans. (A,D)

Sol. $\Delta\phi = (2n+1)\,\lambda/2$

12. Ans. (B, C)

Sol.



$$v = \frac{fu}{f-u} \text{ on the left of lens}$$

$$d = 2\left(\frac{t}{2}\right)\left(\frac{v}{u}-1\right) = t\left(\frac{u}{f-u}\right)$$

$$D = L + \frac{uf}{f-u}$$

Fringe width

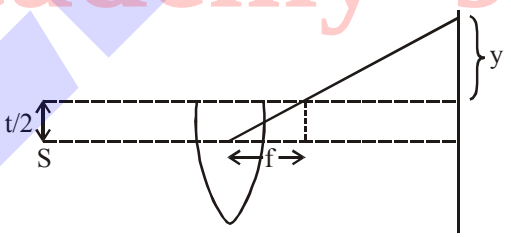
$$B = \frac{\lambda\left(L+\frac{uf}{f-u}\right)}{t\left(\frac{u}{f-u}\right)} = \frac{\lambda}{t}\left[f + \frac{L(f-u)}{u}\right]$$

$$\simeq \frac{\lambda f}{t} = \frac{6\times10^{-7}\times0.4}{10^{-3}}$$

$= 0.24$ mm

$$\frac{y}{L-f} = \frac{t}{2f} \Rightarrow y = \frac{t(L-f)}{2f}$$

Length of interference pattern



$$\frac{t(L-f)}{f} + t$$

$$= t\left(\frac{L}{f}-1+1\right)$$

$$= t\frac{L}{f}$$

$$= \frac{(10-3)\times50}{40}$$

$$= \frac{5}{4\times10^{3}}\text{m} = \frac{5}{40}\text{cm} = \frac{1}{8}\text{cm}$$

13. Ans. (A,B,C)

Sol. $d = 2a(\mu-1)\alpha$

$$\beta = \frac{D\lambda}{d} \Rightarrow \frac{D\lambda}{2a(\mu-1)\alpha} = \beta$$

(A) a is decreased so $\beta \uparrow$

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(B) $\beta' = \dfrac{D(\lambda/\mu_m)}{2a\alpha\left(\dfrac{\mu}{\mu_m}-1\right)} = \dfrac{D\lambda}{2a\alpha}\dfrac{1}{(\mu-\mu_m)}$

Now, $(\mu-\mu_m) < (\mu-1)$
so, $\beta \uparrow$

(C) $\alpha \downarrow \Rightarrow \beta\uparrow$

14. Ans. (A,B,C)

Sol. $\beta = \dfrac{\mu\lambda}{2\theta}$

15. Ans. (B,C,D)

Sol. $\beta = \dfrac{\lambda}{2\mu\theta}$

16. Ans. (A,B)

Sol. $d \sin\theta = n\lambda$

$\sin\theta = \dfrac{n\lambda}{d}$

$d = \lambda \Rightarrow \sin\theta = n$

$n = 0$ only one maxima

$\sin\theta = \dfrac{n\lambda}{d}$

$\lambda < d < 2\lambda$

$n = 0, 1$

$I_{max} = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$; $I_{min} = \left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

SECTION-II

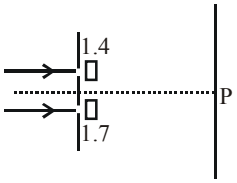
Numerical Answer Type Question **3Q.[3(0)]**

(upto second decimal place)

17. Ans. 9.3 μm

Sol. $\dfrac{3}{4} = \cos^2\dfrac{\Delta\phi}{2}$

$\Delta\phi = 2n\pi + \dfrac{\pi}{3}$ $\left(n+\dfrac{1}{2}\right)\lambda$



5th max & 6th min

phase diff : 10π & 11π

$\dfrac{2\pi}{\lambda}(\mu_2-\mu_1)t = \Delta\phi = 10\pi + \dfrac{\pi}{3}$ is acceptable

$t = \dfrac{31\pi}{3}\dfrac{x\lambda}{x2\pi\times(\mu-\mu_1)} = \dfrac{31}{3}\times 5400\times 10^{-10}$

18. Ans. 1.25

Sol. $\Delta X = 2\ \mu t$

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For destructive interference

$2\mu t = \lambda/2$

$$\mu = \frac{\lambda}{4t} = 1.25$$

19. **Ans.** 0.2 mm

Sol. $0.75\, I_{max} = I = I_{max} \cos^2 \frac{\Delta\phi}{2}$

$$\cos\frac{\Delta\phi}{2} = \frac{\sqrt{3}}{2}$$

$$\frac{\Delta\phi}{2} = \frac{\pi}{6}$$

$$\frac{2\pi}{\lambda} d\sin\theta = \frac{\pi}{3}$$

$$\frac{d(\Delta y)}{D} = \frac{1}{6}(\lambda)$$

$$\Delta y = \frac{D\lambda}{6d}$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **1 Q. [4 M (0)]**

20. **Ans.** 5

Sol. When source is at infinite

Path difference = $S_1P - S_2P$

$$\sqrt{D^2 + d^2} - D = \frac{1}{2}\frac{d^2}{D}$$

For maxima $\frac{1}{2}\frac{d^2}{D} = n_1\lambda$

$$\frac{1}{2} \times \frac{4\times10^{-6}}{\frac{8}{5}m} = n_1 \times 5 \times 10^{-7}$$

$\Rightarrow n_1 = 2.5$

$$\left(S_2S_1 + \sqrt{D^2 + d^2}\right) - S_2P$$

$$\Rightarrow 2\times10^{-3} + \frac{1}{2}\frac{d^2}{D}$$

For max $2\times10^{-3} + \frac{1}{2}\frac{d^2}{D} = n_2 \times 5 \times 10^{-7}$

$\Rightarrow n_2 = 4002.5$

i.e. number of miximas = $n_2 - n_1$

$= 4000 = 3995 + n$

$n = 5$

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SECTION-IV

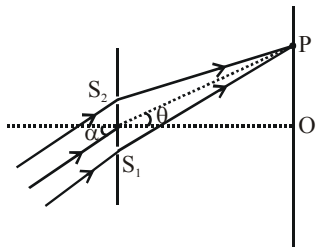
Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

21. Ans. (A)R (B) T (C) QR (D) PS

Sol. Path difference remains same on a circle for case D
Shape of fringe pattern for pin hole is hyperbolic
Shape of fringe pattern for slit is straight line
Δx_{max} can't be greater than 'd' distance between the source in A, B & C & Δx_{min} can't be less than d-distance between the source in D.

22. Ans. (A) (PRST); (B) (QRST); (C) (R); (D) (PRST)

Sol. (A)



$S_1P - (S_2P + d \sin \alpha) = n\lambda$
$d\sin\theta - d \sin \theta = 0$
$\theta = \alpha$
After introduction of sheet
$-(\mu - 1)t + d \sin \theta - d \sin \alpha = 0$
when $\theta = 0$, $-(\mu - 1)t = d \sin \alpha$
not possible

* $\beta = \frac{D\lambda}{d}$

* $d \sin \theta - d \sin \alpha = (n + \frac{1}{2}) \lambda$: Minima

* Put n = 0 in above equation

$d \sin \theta - d \sin \alpha = \frac{\lambda}{2}$: Least order minima

(B) * $S_1P + (\mu_1 - 1)t - [S_2P + (\mu_2 - 1)t] = n\lambda$
$d\sin\theta + (\mu_1 - \mu_2)t = n\lambda$
$n = 0$
$d\sin\theta = -(\mu_1 - \mu_2)t$
zero order max. lies below O
* $d \sin \theta + (\mu' - 1)t' - (\mu_2 - \mu_1)\, t = 0$
for $\theta = 0°$
$(\mu' - 1)t' = (\mu_2 - \mu_1)t$

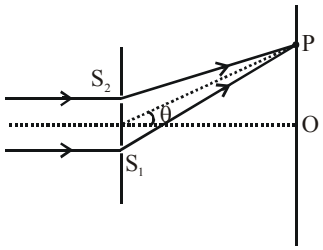
* $\beta = \frac{D\lambda}{d}$

* Point O can be minima
* Point O can be least order minima

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(C)



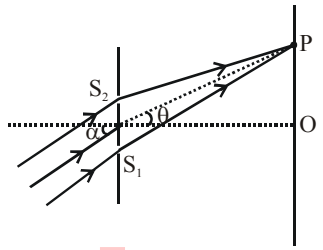
* $S_1P - [S_2P + (\mu - 1)t] = n\lambda$

$d\sin\theta - (\mu - 1)t = 0$: CBF

$$\sin\theta = \frac{(\mu-1)t}{d} \text{ : CBF}$$

$$* \ \beta = \frac{D\lambda}{d}$$

(D)



23. Ans. (A) - (P); (B) - (R); (C) - (P); (D) - (S)

Sol. Path difference for

$$\Delta x_{AB} \frac{2d^2}{D} \qquad \Delta x_{AC} \frac{9d^2}{2D}$$

(A) If $\frac{d^2}{D} = \lambda$ then

$$\Delta x_{AB} = 2\lambda \qquad \Delta x_{AC} = \frac{9}{2}\lambda$$

$$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times 2\lambda \qquad \Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{9}{2}\lambda = 9\pi$$

$= 4\pi$



$\boxed{I_R = I_0}$

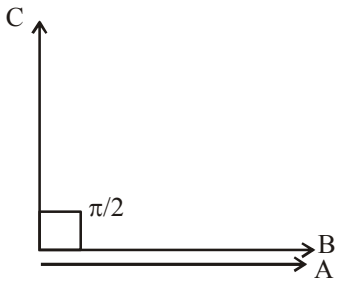
(B) If $\frac{d^2}{D} = \frac{\lambda}{2}$

$$\Delta x_{AB} = 2 \times \frac{\lambda}{2} = \lambda \qquad \Delta x_{AC} = \frac{9}{2} \times \frac{\lambda}{2} = \frac{9\lambda}{4}$$

$$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times \lambda = 2\pi \quad \Delta\phi_{AC} = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times \frac{9\lambda}{4} = \frac{9\pi}{2}$$

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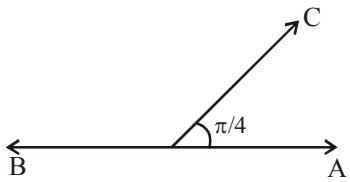


$I_R = 5I_0$

(C) If $\frac{d^2}{D} = \frac{\lambda}{4}$

$\Delta x_{AB} = 2 \times \frac{\lambda}{4} = \frac{\lambda}{2}$ $\Delta x_{AC} = \frac{9}{2} \times \frac{\lambda}{4} = \frac{9\lambda}{8}$

$\Delta\phi_{AB} = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$ $\Delta\phi_{AC} = \frac{2\pi}{\lambda} \times \frac{9\lambda}{8} = \frac{9\pi}{4}$

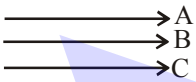


$I_R = I_O$

(D) If $\frac{d^2}{D} = 2\lambda$

$\Delta x_{AB} = 2 \times 2\lambda = 4\lambda$ $\Delta x_{AC} = \frac{9}{2} \times 2\lambda = 9\lambda$

$\Delta\phi = \frac{2\pi}{\lambda} \times 4\lambda = 8\lambda$ $\Delta\phi_{AC} = \frac{2\pi}{\lambda} 9\lambda = 18\pi$

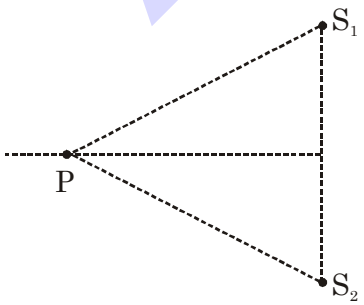


$I_R = 9I_O$

Subjective Type **5 Q. [4 M (0)]**

24. **Ans.** $4I_0$, $\theta = \sin^{-1}\left(\frac{2n+1}{8}\right)$ n = 0, 1, 2, 3, (iii) 393.75 m

Sol. (i) For $\theta = 0$



Suppose plane is at P at any moment.
Then, $PS_1 = PS_2$

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$\therefore$ There is no phase difference between the waves reaching plane from sources S_1 & S_2

$\therefore$ $I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos(\Delta\phi)$

$I = 4I_0$ (as $\Delta\phi = 0$)

for $\theta = \frac{\pi}{2}$

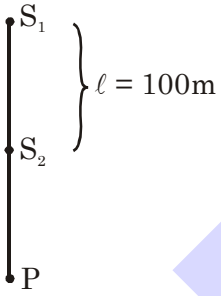
Path difference,

$\Delta x = \ell$

$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$

$= \frac{2\pi f}{C} \times \Delta x$

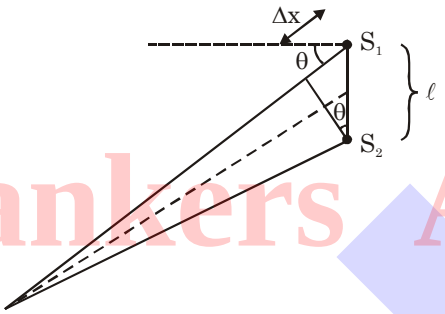
$\Delta\phi = \frac{2\pi \times 12 \times 10^6}{3 \times 10^8} \times 100$



$\Delta\phi = 8\pi \Rightarrow$ waves reach the plane in same phase

Again $I = 4I_0$.

(ii)



Path difference, $\Delta x = \ell \sin\theta$

Aircraft will not receive signal when

$\Delta = (2n+1)\frac{\lambda}{2}$

$\ell \sin\theta = (2n+1)\frac{\lambda}{2}$

$\sin\theta = \left(\frac{2n+1}{2\ell}\right)\lambda$

$\sin\theta = \frac{2n+1}{100} \times \frac{C}{f}$

$\sin\theta = \frac{2n+1}{8} \Rightarrow \theta = \sin^{-1}\left(\frac{2n+1}{8}\right)$

Where n = 0, 1, 2, 3, ...

25. **Ans.** (i) 6 mm, (ii) $50\pi/3$

Sol. (i) $\beta = \frac{\Delta\lambda}{d} = \frac{1 \times 6000 \times 10^{-10}}{0.1 \times 10^{-3}} = 6 \times 10^{-3}$ m

= 6 mm

(ii) Path difference,

$\Delta x = d\sin\theta$

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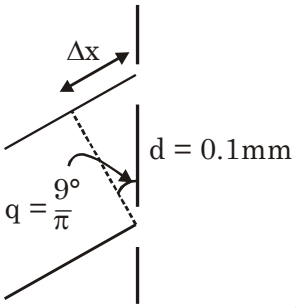
$\Delta x = 0.1 \times \theta$

as θ is small

$$\theta = \frac{9^\circ}{\pi} = \frac{9}{\pi} \times \frac{\pi}{180} \text{rad}$$

$$\theta = \frac{1}{20} \text{rad}$$

$$\Delta x = 0.1 \times 10^{-3} \times \frac{1}{20} = \frac{1}{2} \times 10^{-5}$$



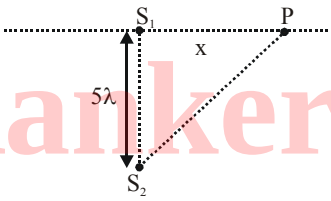
$\therefore$ Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$

$$= \frac{2\pi}{6000 \times 10^{-10}} \times \frac{1}{2} \times 10^{-5}$$

$$= \frac{50\pi}{3}$$

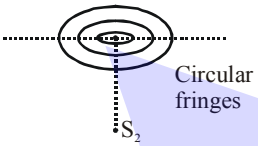
26. **Ans.** 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 m.m.;

Sol.



$S_2P - S_1P = n\lambda$

$$\sqrt{(5\lambda)^2 + x^2} - x = n\lambda$$

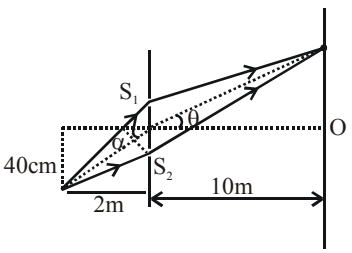


27. **Ans.** (i) y = 2 cm, (ii) m = 1.0016

Sol. $d \sin\theta - d \sin\alpha = 0$

$\theta = \alpha$

$$\tan\theta = \tan\alpha = \frac{40}{200} = \frac{1}{5}$$



$$y = \frac{1}{5} \times 10 = 2\text{cm}$$

$(\mu - 1)t = d \sin\alpha$

$$\mu = \frac{d \sin\alpha}{t} + 1$$

28. **Ans.** (i) 1 mm (ii) increase

Sol. $OA = y = \frac{3D\lambda}{d}$

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$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=\frac{1}{f}+\frac{1}{u}=\frac{1}{10}-\frac{1}{15}=\frac{1}{30}$$

$v = 30$ cm

$D = 1.30 - 0.30$

$D = 1.00$ m

$$d_0 = m = \frac{v}{u}=\frac{30}{15}=2$$

$d_0 = 2 \times 0.5$ mm $= 0.5$ mm

$d = 0.5 + 2d_0 = 1.5$ mm

$$OA = \frac{3\times1\times5\times10^{-7}}{1.5\times10^{-3}}$$

$OA = 10 \times 10^{-4}$

$OA = 1$mm

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